

Noise Models in Digital Image Processing

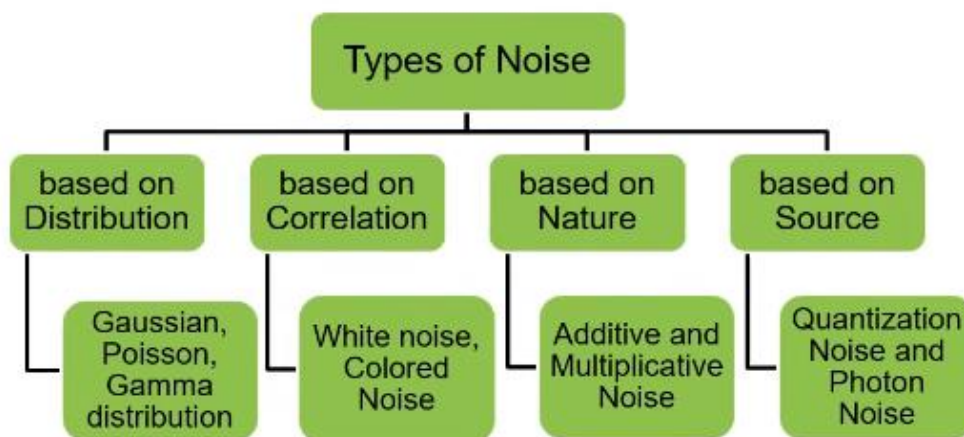
The principal source of noise in digital images arises during image acquisition and transmission. The performance of imaging sensors is affected by a variety of environmental and mechanical factors of the instrument, resulting in the addition of undesirable noise in the image. Images are also corrupted during the transmission process due to non-ideal channel characteristics.

Generally, a mathematical model of image degradation and its restoration is used for processing. The figure below shows the presence of a degradation function $h(x,y)$ and an external noise $n(x,y)$ component coming into the original image signal $f(x,y)$ thereby producing a final degraded image $g(x,y)$. This part composes the degradation model. Mathematically we can write the following :

Where * indicates convolution in the spatial domain.

Noise Modelling

Some of the types of frequent occurred noise that are encountered in Image Processing can be categorized into four categories.



Types of Noise based on Distribution

- ☐ Noise is a fluctuation in pixel values and it is characterized by random variable
- ☐ A random variable probability distribution is an equation that links the values of the statistical result with its probability of occurrence
- ☐ Categorization of noise based on probability distribution is very popular

1. Gaussian Noise

- Random noise that enters a system can be modelled as a **Gaussian** or **normal distribution**
- Gaussian noise affects both dark and light areas of image

The PDF of a Gaussian random variable, z , is given by:

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-\mu)^2 / 2\sigma^2}$$

- Where z represents gray level
- μ is the **mean of average** values of z
- σ is the standard deviation.
- The standard deviation squared, σ^2 , is known as **variance** of z .

2. Impulse Noise

- It is also known as **Shot Noise**, **Salt and Pepper Noise**, and **Binary Noise**
- It occurs mostly because of **sensor** and **memory problem** because of which pixels are assigned incorrect maximum values

PDF of impulse noise

$$P(z) = \begin{cases} P_a & \text{for } z=a \\ P_b & \text{for } z=b \\ 0 & \text{otherwise} \end{cases}$$

$z \rightarrow a \text{ \& } b \rightarrow \text{grey levels}$
 $P_a = P_b = 0$ **unipolar noise**

4. Exponential Noise

- This type of noise occurs mostly due to the **illumination problems**
- It is present in laser imaging

PDF

$$P(z) = \begin{cases} a e^{-az} & \text{for } z \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Mean $\rightarrow \frac{1}{a}$

Variance $\rightarrow \frac{1}{a^2}$

3. Poisson Noise

- This type of noise manifests as a **random structure** or **texture** in images
- It is very common in **X-ray images**

PDF

$$P(z) = \frac{(np)^z}{z!} e^{-np}$$

$n \rightarrow \text{total no. of pixels}$
 $p \rightarrow \text{Ratio of noise pixels to the total no. of pixels}$

5. Gamma Noise

- This type of noise also occurs mostly due to the **illumination problems**

PDF

$$P(z) = \begin{cases} \frac{a^b \times z^{b-1}}{(b-1)!} e^{-az} & \text{for } z \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

* Mean $\rightarrow \frac{b}{a}$

* Variance $\rightarrow \frac{b}{a^2}$

6. Rayleigh Noise

- ❑ This type of noise is mostly present in **range images**
- ❑ Range images are mostly used in remote sensing applications

$$\text{PDF} \\ P(z) = \begin{cases} \frac{2}{b}(z-a) e^{-\frac{(z-a)^2}{b}} & \text{for } z \geq a \\ 0 & \text{otherwise} \end{cases}$$

$$* \text{ Mean} \rightarrow a + \sqrt{\frac{\pi b}{4}}$$

$$* \text{ Variance} \rightarrow \frac{b(4-\pi)}{4}$$

7. Uniform Noise

- ❑ It is also very popular noise occurs in images where different values of noise are equally probable
- ❑ It occurs because of **Quantization noise**

$$\text{PDF} \\ P(z) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq f(x,y) \leq b \\ 0 & \text{otherwise} \end{cases}$$

$$* \text{ Mean} \rightarrow a+b/2$$

$$* \text{ Variance} \rightarrow (b-a)^2/12$$

Types of Noise based on Correlation

- ❑ Statistical dependence among pixels is known as correlation
- ❑ If a pixel is independent of its neighbouring pixels, it is known as uncorrelated pixel otherwise it is known as correlated pixel

- ❑ Uncorrelated Noise is known as **White Noise**
- ❑ Mathematically for white noise, the **noise power spectrum** or **power spectral density** remains constant with frequency

- ❑ Characterization of colored noise is quite difficult because its origin is mostly unknown
- ❑ One popular colored noise is **Pink Noise**
- ❑ Its **power spectrum** is **not constant**, rather it is proportional to reciprocal of frequency
- ❑ This is also known as **1/f** or **Flicker Noise**

Types of Noise based on Nature

Based on the nature of its presence, noise can be modelled in two ways

1. Additive Noise

- ❑ In this case an image can be perceived as the image plus noise
- ❑ This is a **linear** problem

$$\text{Perceived image,} \\ g(x,y) = f(x,y) + n(x,y) \\ \downarrow \quad \quad \downarrow \\ \text{i/p image} \quad \text{noise}$$

2. Multiplicative Noise

- ❑ It can be modelled as multiplicative process
- ❑ Speckle noise is most encountered multiplicative noise in image processing
- ❑ It is mostly present in **medical images**
- ❑ It can be modelled as pixel value multiplied by the random value

$$I = f(x,y) + [f(x,y) \times N_g]$$

Random noise having a zero mean gaussian PDF

Types of Noise based on Source

Noise based on source, commonly encountered in image processing are:

- ☐ Quantization Noise
- ☐ Photon Noise

☐ Quantization Noise occurs due to a **difference between** the actual and allocated values

☐ It is **inherent** in the quantization process

☐ Photon Noise occurs due to the statistical nature of electromagnetic waves

☐ Generation of photon is **not constant** because of statistical variation

☐ This causes variation in **photon count** which is known as photon noise

☐ It is present in many **medical images**

Estimation of Degradation Function

Degradation function can be estimated by three principal methods, mentioned below:

Estimation by Observation

Estimation by Experimentation

Estimation by Modelling

Estimation of Degradation Function

Estimation by Observation

- ✓ The degradation function (H) can be estimated by **visually looking into a small section of the image**
- ✓ Given a small **sub image** $g_s(x,y)$, we can manually (i.e. filtering) remove the degradation in that region with an estimated sub image
- ✓ Let us assume that the additive noise is negligible in such an area with a strong signal content

$$H_s(u,v) = \frac{G_s(u,v)}{\hat{F}_s(u,v)}$$

$H_s(u,v)$ estimated for such a small sub image, the shape of this degradation function can be used to get an estimation of $H(u,v)$ for the entire image

Estimation by Experimentation

- ❑ Suppose that **equipment** similar to the one used for acquisition be available, then it is possible to obtain an **accurate estimation** of the degradation by using the same device
- ❑ This can be achieved by applying an impulse (bright dot) as an input image
- ❑ The Fourier transform of an impulse is constant, therefore

$$H(u,v) = \frac{G(u,v)}{A}$$

Where,

- ✓ A is a constant describing the strength of the impulse
- ✓ The effect of noise on an impulse is negligible here

Estimation by Modelling

- ❑ A model is a **set of equations** that approximate a real system
- ❑ Many types of models have been developed for **modelling degradations** like motion blurs and atmospheric turbulence

Two types of techniques are available,
based on the estimation of blurring function

Direct Estimation
Technique

Indirect Estimation
Technique

Direct Estimation Technique

- Blur function is directly measured by isolating the image of an object pixel
- By observing the point source object, we may get the impulse response of blur because it is the image of point source object
- If we take many such samples and by observation and estimation, a covariance function can be estimated for a relatively uniform region of the image

Indirect Estimation Technique

- Many techniques can be formulated. Let us discuss one technique
- Degradation model that has been constructed is the convolution of the image $f(x,y)$, blurring function $h(x,y)$ and additive noise
- Assume that there is no noise

$$g(x,y) = f(x,y) \times h(x,y) \text{ (Spatial domain)}$$

$$G(u,v) = F(u,v)H(u,v) \text{ (Freq. Domain)}$$

$$\ln[G(u,v)] = \ln[F(u,v)] + \ln[H(u,v)]$$

$M \rightarrow$ Dimension of image

Possible estimation of $H \rightarrow$

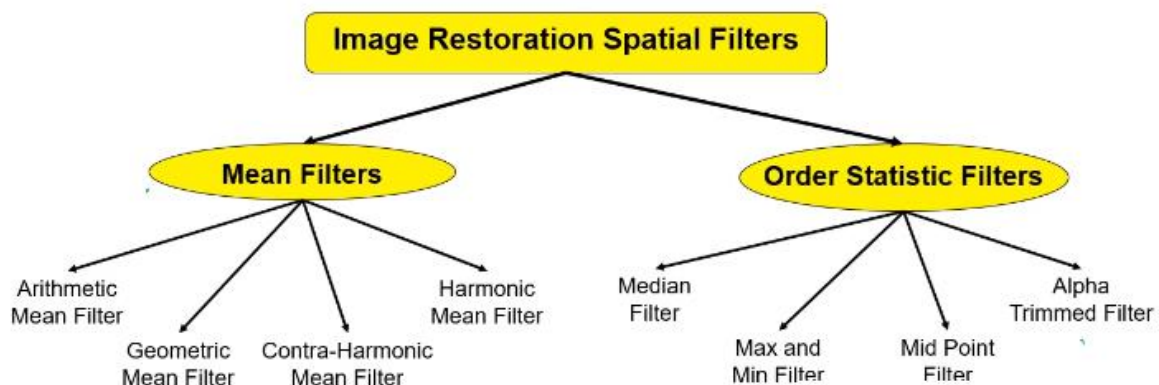
$$\log H = \frac{1}{M} \left[\sum_{k=1}^M [\log |v_k| - \log |u_k|] \right]$$

Example: some portion of relatively good part of the image (u_k) and the degraded image (v_k) can be obtained

H
statistical
means

Image Restoration in presence of Noise Only

Spatial Filter, which are used in image smoothing and sharpening, are also useful for removing noise. Classification is shown as:



Mean Filters

1. Arithmetic Mean Filter

- ❑ This filter removes local variations within the image
- ❑ It is similar to **low pass filter**
- ❑ It is useful in removing **Gaussian Noise** and **Uniform Noise**

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(s,t) \in S_{x,y}} g(s, t)$$

2. Geometric Mean Filter

- ❑ This filter eliminates **Gaussian Noise**
- ❑ It is **ineffective** for Pepper type of Noise
- ❑ Geometric mean filter achieves smoothing comparable to the arithmetic mean filter, but it tends to lose less image detail in the process

$$\hat{f}(x, y) = \left[\prod_{(s,t) \in S_{x,y}} g(s, t) \right]^{\frac{1}{mn}}$$

Mean Filters

3. Harmonic Mean Filter

- ❑ It works well for **salt noise**, but fails for pepper noise.
- ❑ It does well also with other types of noise like **Gaussian noise**

$$\hat{f}(x, y) = \frac{mn}{\sum_{(s,t) \in S_{x,y}} \frac{1}{g(s, t)}}$$

4. Contra-harmonic Mean Filter

- ❑ It is well suited for reducing the effects of **salt-and-pepper noise**
- ❑ $Q > 0$ for elimination of pepper noise and $Q < 0$ for elimination of salt noise
- ❑ Q is the **order of the filter**

$$\hat{f}(x, y) = \frac{\sum_{(s,t) \in S_{x,y}} [g(s, t)]^{Q+1}}{\sum_{(s,t) \in S_{x,y}} [g(s, t)]^Q}$$

Order Statistic Filters

- ❑ Order statistic filters are also known as **rank, rank order or order filters**
- ❑ These filters are not based on convolution
- ❑ These filters are differentiated based on how they choose the values in the sorted list. The position indicates the rank.

1. Median Filter

- ❑ It is an example of **non linear filter**
- ❑ It simply sorts the list and finds the median
- ❑ Centre pixel is replaced by median value

$$f(x, y) = \text{median}_{(s,t) \in S_{x,y}} \{g(s, t)\}$$

2. Maximum Filter

- ❑ This filter selects the largest value in the sorted list
- ❑ It is used for removing **pepper type of noises**

$$f(x, y) = \max_{(s,t) \in S_{x,y}} \{g(s, t)\}$$

Order Statistic Filters

3. Minimum Filter

- ❑ This filter selects the smallest value
- ❑ It is very effective in elimination of **salt type of noises**

$$f(x, y) = \min_{(s,t) \in S_{xy}} \{g(s, t)\}$$

4. Midpoint Filter

- ❑ This filter selects the midpoint
- ❑ Midpoint is the average of minimum and maximum values
- ❑ It is very effective in removing **Gaussian noise** and **Uniform noise**

$$f(x, y) = \frac{1}{2} \left[\max_{(s,t) \in S_{xy}} \{g(s, t)\} + \min_{(s,t) \in S_{xy}} \{g(s, t)\} \right]$$

Order Statistic Filters

5. Alpha-trimmed Mean Filter

- ❑ It is based on the **concept of computation** of the average of the pixels that falls within the window
- ❑ It is quite similar to simple **average filter** but the difference is that user can clip some of the pixels by specifying an alpha value
- ❑ This filter is useful for removing **Gaussian noise** and **Salt and Pepper noise**

Selection of the pixel value for replacing the central pixel value is represented by the following formula:

$$f(x, y) = \frac{1}{mn - d} \sum_{(s,t) \in S_{xy}} \{g_r(s, t)\}$$

- ✓ We delete the d/2 lowest and d/2 highest intensity values of the g(s,t) in the neighbourhood S_{xy}
- ✓ Let $g_r(s, t)$ represents the remaining (mn-d) pixels

Periodic Noise

- ❑ Periodic Noise is **sinusoidal** at multiples of a specific frequency and **periodic in nature**
- ❑ The occurrence of **uniform bars** over an image is a manifestation of periodic noise
- ❑ It typically arises from **electrical** or **electromechanical interference** during image acquisition, and is **spatial dependent**

Estimation of Periodic Noise Parameters

- ❑ The parameters of periodic noise are typically estimated by inspection of the Fourier spectrum. The frequency spikes can often be detected by visual analysis.
- ❑ Automatic analysis is possible if the noise spikes are either exceptionally pronounced or a priori knowledge about their location is known.

Periodic Noise can be removed by Band-pass, Band-reject and Notch Filter

Band-Pass Filters

Band-pass Filters allow frequencies within a particular range to pass through and attenuate all other frequencies

- This filter allows the frequencies if they fall in the range $D_l - D_h$
- This range is known as Band

1D Transfer Function is given by:

$$H(D) = \begin{cases} 1 & \text{for } D_l \leq D \leq D_h \\ 0 & \text{for } D > D_0 \end{cases}$$

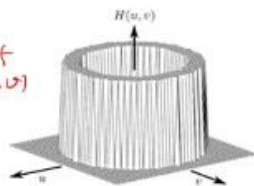
D₀ → cut-off freq. or 1.D

The transfer function for 2D band-pass filter is given as follows:

$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) < D_0 - \frac{W}{2} \\ 1 & \text{if } D_0 - \frac{W}{2} \leq D(u,v) \leq D_0 + \frac{W}{2} \\ 0 & \text{if } D(u,v) > D_0 + \frac{W}{2} \end{cases}$$

D(u,v)
↓
distance of the point (u,v) from the center

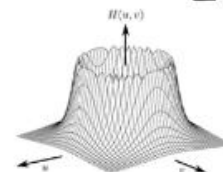
W → width of band



Ideal Band-pass Filter

A Butterworth band-pass filter of order n is given as :

$$H_{bp}^{Butterworth}(u,v) = \frac{\left[\frac{D(u,v) \omega}{D^2(u,v) - D_0^2} \right]^{2n}}{1 + \left[\frac{D(u,v) \omega}{D^2(u,v) - D_0^2} \right]^{2n}}$$



Butterworth Band-pass Filter

Band-Reject Filters

The objective of Band-reject Filter is to attenuate limited range Frequencies while leaving the other frequencies

Band-reject Filter is the complement of a band pass filter

These filters are very effective in removing periodic noise and ringing effect is normally small

$$H_{br}(u, v) = 1 - H_{bp}(u, v)$$

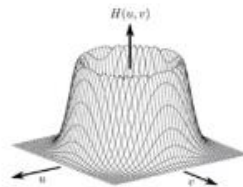
This filter rejects the frequencies if they fall in the range $D_l - D_h$

1D Transfer Function is given by:

$$H(D) = \begin{cases} 0 & \text{for } D_l \leq D \leq D_h \\ 1 & \text{for } D > D_0 \end{cases}$$

A Gaussian band-pass filter is given as follows:

$$H_{bp}^{Gaussian}(u, v) = e^{-\frac{1}{2} \left[\frac{D^2(u, v) - D_0^2}{D(u, v) \omega} \right]^2}$$



Gaussian Band-pass Filter

An ideal band reject filter is given as follows:

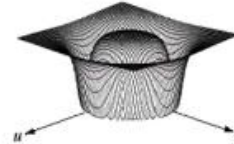
$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) < D_0 - \frac{W}{2} \\ 0 & \text{if } D_0 - \frac{W}{2} \leq D(u, v) \leq D_0 + \frac{W}{2} \\ 1 & \text{if } D(u, v) > D_0 + \frac{W}{2} \end{cases}$$



Ideal Band Reject Filter

A Butterworth band reject filter of order n is given as :

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v)W}{D^2(u, v) - D_0^2} \right]^{2n}}$$



Butterworth Band Reject Filter
(of order 1)

A Gaussian band reject filter is given as follows:

$$H(u, v) = 1 - e^{-\frac{1}{2} \left[\frac{D^2(u, v) - D_0^2}{D(u, v)W} \right]^2}$$



Gaussian Band Reject Filter

Image Compression Model

Image compression is very important task in image processing
Images and Videos require lots of space and large transmission time

Data compression is the process of encoding data
so that it takes less storage space or less transmission
time than it would if it were not compressed

Data Compression:

It is the Mathematical process
of transforming data to a smaller
representation from the original



- ☐ If $N1 = N2$ There is no compression
- ☐ If $N2 \ll N1$ There is significant Compression
- ☐ If $N1 \ll N2$ Data Explosion

Image Compression Model

Data and Information are two different things:

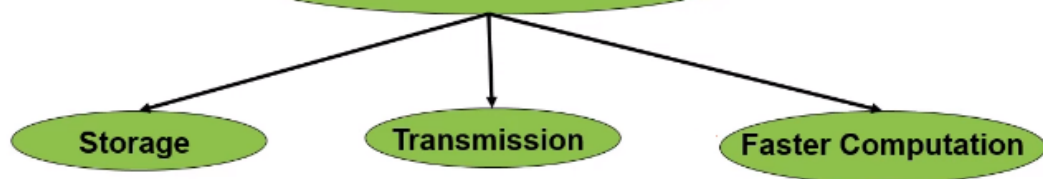
- ☐ Data is raw facts which are encountered in image processing
- ☒ Information is an interpretation of the data in a meaningful way
- ☒ Data is the means by which information is conveyed

Types of Data

Text Data	Binary Data	Image Data	Graphics Data	Sound Data	Video Data
• Data present in flat files • It can be read and understood by human beings	• Machine can interpret it • Ex: metadata present in database files	• This is pixel data • It contains intensity and color information of image	• This data is in vector form	• This is audio information	• This represents video information

Image Compression Model

Why Data Compression?



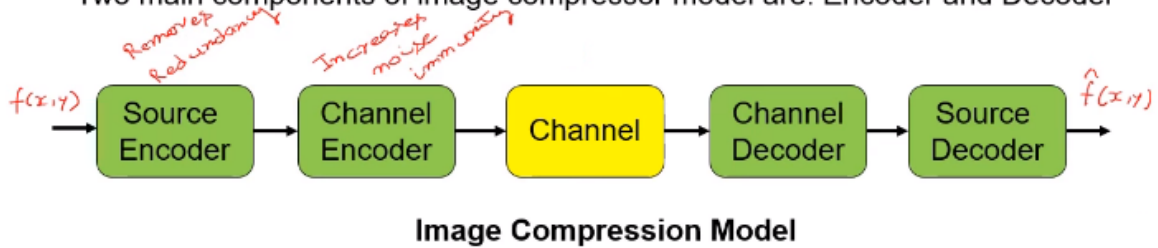
Applications of Data Compression:

- ☒ Personal communication like Facsimile, Voice mail and telephony
- ☐ Computer networks – Internet
- ☐ Multimedia applications
- ☐ Image and signal processing
- ☐ Digital and Satellite TV
- ☐ Video conferencing and Digital Library

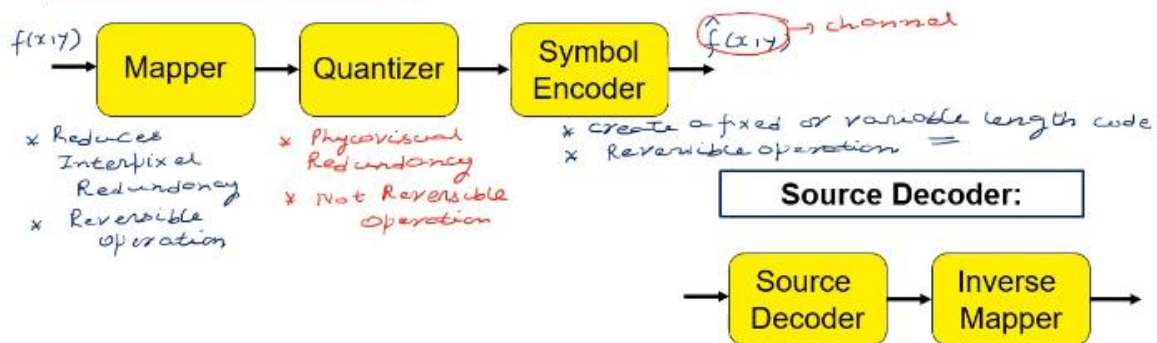
Image Compression Model



Two main components of image compressor model are: Encoder and Decoder



Stages of encoding:



Compression Measures

Data Compression:

It is the Mathematical process of transforming data to a smaller representation from the original

EX:-

JPEG

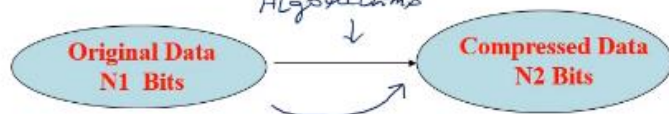
Joint Photographic Experts group

✓ → Logical compression

✓ → Physical compression

$$\text{Compression Ratio (CR)} = \frac{N_1}{N_2}$$

Compression Algorithms



$$\text{Relative Redundancy (RD)} = 1 - \frac{1}{CR}$$

Data Compression algorithms can be visualized as the mapping of a set of message symbols to codes using some logical rules of conversation

✓ Pixels or block of Pixels → set of symbols

✓ Code → sequence of symbols or numbers

✓ Code word → string of codes



- ❑ If $N_1 = N_2$ There is no compression
- ❑ If $N_2 \ll N_1$ There is significant Compression
- ❑ If $N_1 \ll N_2$ Data Explosion

$$CR = \frac{N_1}{N_2} \rightarrow \infty$$

$$RD = 1$$

Compression is done

$N_1 < N_2$ (Reverse compression) x

$$CR = \frac{N_1}{N_2} \rightarrow 0 \quad RD = 1 - \frac{1}{CR} \rightarrow \infty$$

① $N_1 = N_2$
 $CR = \frac{N_1}{N_2} = 1$
 (compression ratio)

Relative Redundancy $(RD) = 1 - \frac{1}{CR}$
 $= 0$
 i/p message is same as o/p message

Data in Transformed Set > Original Data

Some of the Metrics used to quantify compression measures are:

Compression Ratio

(CR)

$$\text{Compression Ratio} = \frac{\text{Uncompressed file size}}{\text{Compressed file size}} = \frac{N_1}{N_2}$$

Saving Percentage

(SP)

$$\text{Saving Percentage} = 1 - \left(\frac{\text{Compressed file size}}{\text{Uncompressed file size}} \right) = 1 - \frac{N_2}{N_1}$$

Bit Rate

BR → indicates efficiency of compression Algorithm

$$\text{Bit Rate} = \frac{\text{Size of Compressed File}}{\text{Number of pixels in Image}} = \frac{N_2}{N}$$

bps
kpbs
Mbps

Example: original image is of size 256 x 256 pixels, gray-scale i.e. 8 bits/pixel, file is 65536 bytes (64K). After compression image file is of 6554 bytes

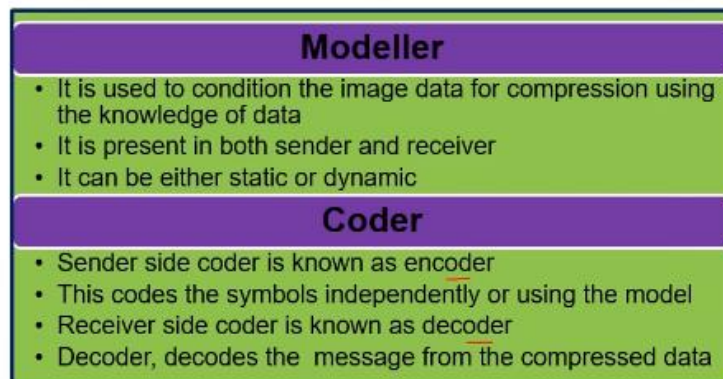
$$\begin{aligned} \text{Compression Ratio} &= N_1 / N_2 \\ &= 65536 / 6554 \\ &= 10 \\ &= 10:1 \\ &= 10 \text{ to } 1 \text{ compression} \\ &= 10 \text{ times compression} \end{aligned}$$

It means image has been compressed to 1/10 original size

Compression Algorithms and Its Types

The objective of **Compression Algorithm** is to reduce the source data to a compressed form and decompress it to get the original data

Any Compression Algorithms has two components:



Modeller

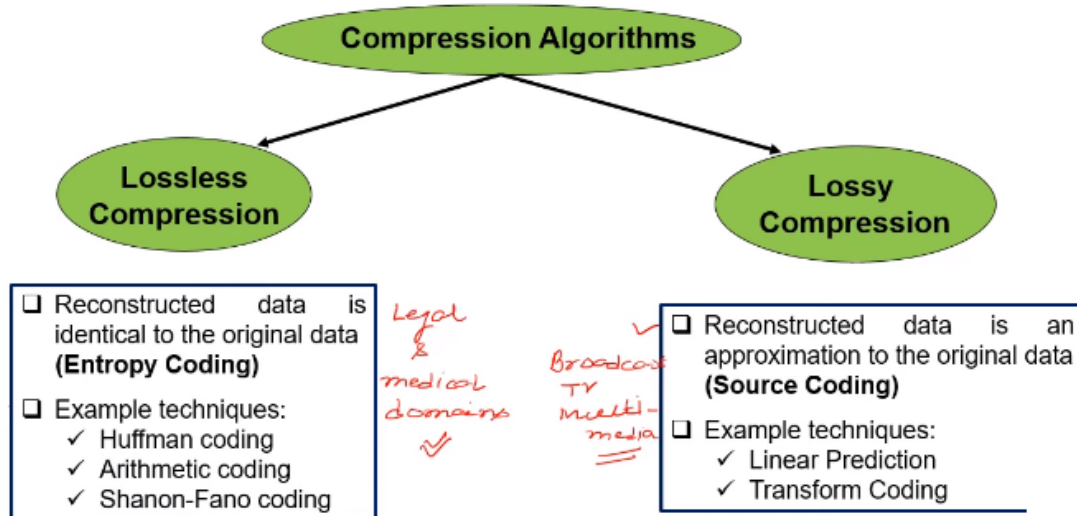
- It is used to condition the image data for compression using the knowledge of data
- It is present in both sender and receiver
- It can be either static or dynamic

Coder

- Sender side coder is known as encoder
- This codes the symbols independently or using the model
- Receiver side coder is known as decoder
- Decoder, decodes the message from the compressed data

✓ Static
do not change
✓ Dynamic
↓
models change depending upon the change of data

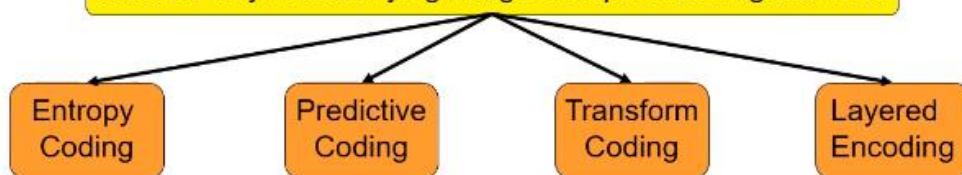
Compression Algorithms and Its Types



Difference between Lossless and Lossy Compression

Lossless Compression	Lossy Compression
Reversible Process	Non-reversible Process
No information is lost	Some information is lost
Compression Ratio is less	Compression Ratio is very high
It is used for that data which human can handle directly like Text data	It is used for diffused data which human can not understand or interpret
Compression is independent on psychovisual system	Compression is dependent on psychovisual system

Another way of classifying Image Compression Algorithm is:



Entropy Coding

- The average information in an image is known as its entropy
- Coding is based on the entropy of the source and on the possibility of occurrence of the symbols
- ✓ An event that is less likely to occur is said to contain more information than an event that is more likely to occur

Set of symbols (alphabet) $S = \{s_1, s_2, \dots, s_N\}$,
 N is number of symbols in the alphabet
 Probability distribution of the symbols: $P = \{p_1, p_2, \dots, p_N\}$

According to Shannon, the entropy H of an information source S is defined as follows:

$$H = - \sum_{i=1}^N p_i \cdot \log_2(p_i)$$

Predictive Coding

- ❑ The idea is to remove the **mutual dependency** between the **successive pixels** and then perform the encoding
- ❑ Normally the **samples** would be very **large** but the **difference** would be **small**

Ex.

Ex.

Pixels	→	400	405	420	430
Difference	→		5	15	10

(Note: In the original image, red checkmarks are placed below the difference values 5, 15, and 10.)

ex.

300	4096	120	5081	10
└──────────┘		└──────────┘		
		xx		

Transform Coding

- ❑ Objective is to exploit the **information packing capability** of the transform
- ❑ Energy is packed into **fewer components** and only these components are encoded and transmitted
- ❑ Idea is to **remove the redundant high frequency components** to create compression
- ❑ Removal of these frequency components leads to loss of information
- ❑ This loss of information, if tolerable, can be used for **imaging and vision applications**

Layered Coding

- ❑ It is very useful in case of **layered images**
- ❑ Data structures like **pyramids** are useful to represent an image in **multiresolution form**
- ❑ Sometimes these images are **segmented** as **foreground** and **background**, and based on the application requirement encoding is done
- ❑ It is also in the form of **selected frequency coefficients** or **selected bits of pixels** in an image